Single-molecule Imaging (Contd.)

Lec 8-1

Topics for today

1.) Review of PSF : width, pixel sampling, localization uncertainty resolution.

$s/\sqrt{N}$.

2.) Specimen plane ; weighing background rejection against signal loss.

3) Blinking of fluorophore $\Rightarrow$ STORM : reconstruct structures from sparse events

4) STED : how stimulated emission suppresses fluorescence and narrows the responding region.

5.) MINFLUX : how an excitation minimum encodes position in MINFLUX count ratios.

6.) Fluorescence $\leftrightarrow$ force

$\updownarrow$ $\nwarrow\!\searrow$ mechanical compliance

relative molecular coordinate

7). Likelihood, switching, pulsed STED, and correlated thermal motion

Notations for today

X_0 = true position of fluorophore

$\hat{X}_0$ = Estimated position

s = std of Gaussian PSF

σ_{loc} = uncertainty in estimated center of Gaussian PSF

Imaging one fluorophore

Assume 1D:

x_0 = fluorophore's true coordinate in specimen-plane axis

X_i = random landing coordinate of photon i

x_i = value we observe.

$$X_i = x_0 + \varepsilon_i \qquad \langle \varepsilon_i \rangle = 0 \qquad Var(\varepsilon_i) = s^2$$

↑ displacement produced by PSF

} Neglecting background photons

For Gaussian PSF: $h(x - x_0) = \frac{1}{\sqrt{2\pi}\, s} \exp\left[-\frac{(x - x_0)^2}{2s^2} \right]$

↑ Prob. Density function (PDF)

$h(x - x_0)\, dx$ = probability of a photon landing in $(x, x + dx)$.

so $\int_{\text{everywhere}} h(x - x_0)\, dx = 1$ ← normalization

Let $N \equiv$ # of photons collected while fluorophore is perfectly still. (not moving during the light exposure).

Then centroid estimator:

$$\hat{X}_0 = \frac{1}{N} \sum_{i=1}^{N} X_i$$

Then

$$\langle \hat{X}_0 \rangle = \left\langle \frac{1}{N} \sum_{i=1}^{N} X_i \right\rangle = \frac{1}{N} \sum_i \langle X_i \rangle$$

$$= \frac{1}{N} \sum_{i=1}^{N} (x_0 + \langle \varepsilon_i \rangle)$$

$$= x_0.$$

i.e., the estimator is unbiased: repeated estimates average to the true position.

Note that $\hat{X}_0 - x_0 = \frac{1}{N} \sum_i (X_i - x_0) = \frac{1}{N} \sum_i \varepsilon_i$

Then $Var(\hat{X}_0) = \langle (\hat{X}_0 - x_0)^2 \rangle = \frac{1}{N^2} \left\langle \left(\sum_i \varepsilon_i \right) \left(\sum_j \varepsilon_j \right) \right\rangle$

$$= \frac{1}{N^2} \sum_{i,j} \langle \varepsilon_i \varepsilon_j \rangle$$

$$= \frac{1}{N^2} \left[\sum_i Var(\varepsilon_i) + 2 \sum_{i<j} \underbrace{cov(\varepsilon_i, \varepsilon_j)}_{=0} \right]$$

$$\Rightarrow \mathrm{Var}(\hat{x}_0) = \frac{1}{N^2} \cdot N s^2$$

$$= \boxed{\frac{s^2}{N}} \Rightarrow \boxed{\sigma_{loc} = \frac{s}{\sqrt{N}}}$$

Note that N = total # of photons collected.
N can be the total collected during one exposure or
the total # of photons combined from multiple exposures.

. key is that the fluorophore remains stationary during the collection of N photons (either during one or multiple exposures).

In FIONA, need to ~~fit~~ have dwell time of the kinesin (motor protein)
(yildiz & selvin)
be not too short
⇒ then can fit multiple exposures ~~both~~ before the kinesin takes the next step.

Key to FIONA:
1) very bright ~~photo~~ & photostable fluorophore
(i.e. can emit lots of photons many times)

2) very good camera & microscope
(i.e. very low detector noise
& a stable, calibrated microscope.)

⇒ Allows for measurement of small changes in photon count across neighboring pixels accurately.

E.g. 100nm = pixel length.
When a fluorophore moves 8 nm,
- pixel on one side gets slightly more photons
- pixel on the other side gets slightly less photons
- Fitting that redistribution reveals the shift of the center.

Contd.

With 100 nm = pixel width, (say $s = 10$mm too.)

$$\sigma_{loc} = \frac{100\,nm}{\sqrt{N}}$$

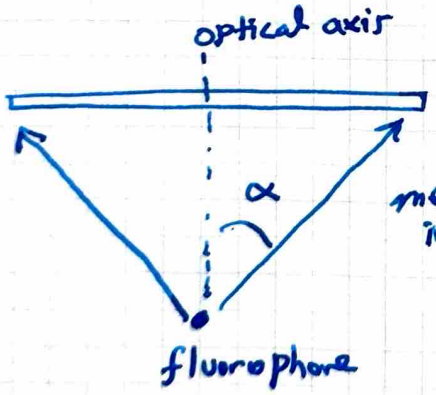
Detected photons per estimate	σ_{loc} = position uncertainty
100	$\Rightarrow$ 10nm
10,000	$\Rightarrow$ 1nm.

$\therefore$ An 8-nm shift would be difficult to establish from two estimates each uncertain by 10nm.

NA = Numerical Aperture : $\boxed{NA = n \sin\alpha}$

↳ describes the range of light directions an object can collect, accounting for the medium through which the light travels. It determines how tightly the microscope can image a point source.

(Higher NA, the better it is).

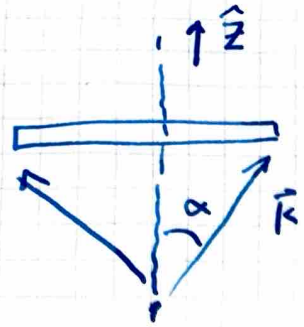


} objective of microscope collects this light cone

Why define NA = n sinα?

Ans: $\lambda_0 \equiv$ vacuum wavelength. n = medium wave length.

$$\lambda_{med} = \frac{\lambda_0}{n} \Rightarrow \underset{\text{wave number}}{k} = \frac{2\pi}{\lambda_{med}} = \frac{2\pi n}{\lambda_0}$$



$k_z = k\cos\theta \quad k_\perp = k\sin\theta$

transverse = $\perp$ to optical axis.

The objective accepts directions only up to $\theta = \alpha$.

the largest accepted transverse wavevector is

therefore $k_{\perp,max} = k\sin\alpha = \frac{2\pi}{\lambda_0} n\sin\alpha$

↳ NA naturally appears.

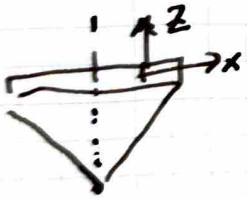
Writing $k_0 = \frac{2\pi}{\lambda_0}$, we have!

$$NA = \frac{k_{\perp,max}}{k_0} = n\sin\alpha.$$

∴ NA is the maximum transverse wavevector the objective collects, expressed in units of the vacuum wavenumber.

Why does the transverse wavevector matter for imaging?

Consider plane wave in the xz-plane (z = optical axis)



$$E(x,z,t) \propto \exp\left(i[k_x x + k_z z - \omega t]\right)$$

At fixed z & t, its phase changes across the specimen as $k_x x$. A larger k_x means more rapid spatial variation along x.

To form a narrow image of a point source, the lens must combine waves with a broad range of these transverse vectors.

This is the same Fourier principle you know from wave packets:

i.e., A broader range of wavevectors allows a narrower spatial distribution.

∴ larger NA → larger range of transverse wavevector → narrower PSF

Contd.

For the ideal circular pupil used in scalar diffraction calculation, the first dark ring ("Airy ring") occurs at radius $r_0 \approx 0.61 \frac{\lambda_0}{NA}$

Einstein's principle of stimulated emission (STED microscopy by Stefan Hell).

Lay Idea: In blackbody radiation, light & matter are in thermal equilibrium.

⇒ photon not only gets absorbed, ~~It~~ it must also stimulate emission of photon.

Namely:

ground-state molecule + one photon → excited molecule.

and Excited molecule + one incoming photon → ground-state molecule + two outgoing photons.

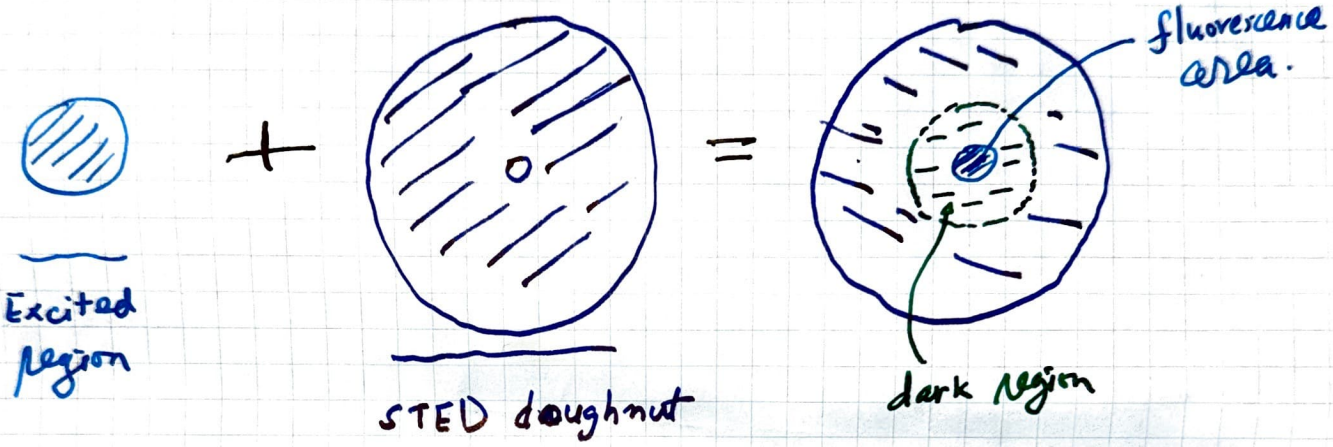
Quantum mechanically:

1) given that light is an oscillating field, light can drive transitions between two energy levels when the light's frequency matches their energy separation: $\hbar\omega = E_{excited} - E_{ground}$.

2) Brighter-intensity light makes stimulated emission happen faster. An excited molecule therefore becomes more likely to release its energy through stimulated emission before it emits spontaneously

In STED microscopy :

the 1st laser puts molecules into excited states.
the doughnut beam then stimulates emission around the edges, adding photons to the STED light.
we filter out that light, leaving the spontaneous fluorescence from near the dark center to form the image.



with the original STED, you can localize fluorophores to ~20 nm resolution.

STORM

Lec 8-8

How sparse must the bright subset be?

$\rho = Mp$ ↑ mean # of ON fluorophores

$K \equiv$ # of ON fluorophores ↑ random variable.

p = probability of turning on.

Poisson: $\boxed{P(K) = \frac{\rho^K e^{-\rho}}{K!}}$

$$P(K \geq 2) = 1 - P(0) - P(1) = 1 - e^{-\rho}(1+\rho)$$

$$P(K \geq 2 \mid K \geq 1) = \frac{1 - e^{-\rho(1+\rho)}}{1 - e^{-\rho}} \simeq \rho/2.$$

If $M = 100$, $p = 0.001$, then: $\rho = 0.1$ ~~or~~ less than 1 molecule ~~ON~~ in average.

and $\rho/2 \sim 5\%$ chance of overlap between fluorophores in a diffraction limited spot.

Optical tweezers

Lec 8-9

Basic Idea: non-light-absorbing bead (so doesn't heat up).
(e.g. dielectric bead).
↑ insulator.

- the electric field of laser polarizes the bead.
Hence the bead can respond to light even though it has no net charge.

- the non-uniform field of laser produces a strong ↓ if focus is tight. restoring force
i.e., for the usual bead with a refractive index higher than the surrounding H_2O, the gradient force pulls it toward higher light intensity (i.e., near focus).
For a bead that's large enough, we can use ray optics to explain: the bead redirects light, changes its momentum, and thus experiences the corresponding reaction force.

(see Ashkin's force calculation)

(Low light absorption limits heating)

Lec 8-10.

Optical tweezer : • ssDNA : flexible
• dsDNA : rigid. (stiffer)

1) At 3pN (low force) : stiffening wins.

At 3 pN, ssDNA region has considerable bending & fluctuations. Its end-to-end distance is smaller than the length you'd measure by following its backbone.

So when the probe binds, it makes that region stiffer.
⇒ The region can now extend farther along the pulling direction

(i.e. Binding → less bending → greater extension)

2) At 10pN : (higher force) : Helix's shorter length wins.

The force has already pulled the single-stranded region straighter. It can extend farther than the corresponding double helix.

Binding organizes that stretched strand into a helix, reducing its extension.

(i.e., Binding → formation of shorter helical segment → smaller extension).